

TEST PLAN

Authentication & Course Management

Educational Platform — API Testing

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Testing Type: Manual API Testing

Status: Completed

Version: 1.0

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1. Introduction

This document describes the test plan for the Educational Platform API (Version 2). It covers the testing strategy, scope, approach, and test cases for the Authentication, Verification, and Course Management modules.

2. Test Objectives

- Verify that all API endpoints function correctly according to requirements.
- Identify and document defects in the Authentication, SendCode, ValidateCode, and Course Management modules.
- Ensure proper validation of input fields and appropriate error messages are returned.
- Validate that unauthorized users cannot access protected endpoints.
- Ensure integration between system components works as expected.

3. Scope

3.1 In Scope

Module	Endpoints Covered
Authentication	POST /api/Accounts/SignUp POST /api/Accounts/Login
Verification	POST /api/Accounts/SendCode POST /api/Accounts/ValidateCode
Course Management (Admin)	GET /api/Courses/GetCourseById POST /api/Courses/CreateCourse PATCH /api/Courses/UpdateCourseById DELETE /api/Courses/DeleteCourseById

3.2 Out of Scope

- Frontend / UI testing
- Performance and Load testing
- Security Penetration testing
- Email delivery verification (handled manually)
- Profile module
- Browse Courses module (Student)

4. Test Approach

All testing was performed using Postman as the primary API testing tool against a locally hosted server (Local Development Environment). Tests were executed manually by sending HTTP requests and validating responses against expected outcomes. Response validation was performed using Postman Tests where applicable to verify expected outcomes.

4.1 Test Techniques

Technique	Description
Equivalence Partitioning	Valid and invalid input groups tested for each field.
Boundary Value Analysis	Phone number length (11 digits), password length (8-100 chars).
Negative Testing	Invalid formats, missing fields, wrong credentials.
Error Guessing	Realistic but invalid values (e.g., 00000000000 as phone number).
Static Review	API design reviewed for security and REST best practices.

5. Testing Levels

5.1 System Testing

End-to-end testing of complete user flows to verify the system behaves correctly as a whole:

- Full registration flow: `SignUp` → `SendCode` → `ValidateCode` → `Login`
- Password reset flow: `SendCode` (`isForResetPassword=true`) → `ValidateCode` → `ResetPassword`
- Admin course management flow: `Login` → `CreateCourse` → `GetCourseById` → `UpdateCourse` → `DeleteCourse`

5.2 Integration Testing

Testing the interaction between API endpoints and the database, and between modules:

- Verify that registered user data is correctly persisted in the database.
- Verify that verification codes generated by `SendCode` are correctly validated by `ValidateCode`.
- Verify that JWT token issued at `Login` is accepted by protected endpoints.
- Verify that course creation via `CreateCourse` is reflected in `GetCourseById`.

5.3 Functional Testing

Testing that each endpoint performs its intended function correctly, including all happy paths and negative scenarios.

5.4 Validation Testing

Testing input field validation rules, including format validation, required field checks, and business rule enforcement.

5.5 Static Testing / Design Review

Review of API design and endpoint structure without executing the code, focusing on security risks, REST best practices, and design consistency. Findings are documented in a separate Review Findings Report.

6. Types of Testing Performed

Testing Type	Applied To	Status
Functional Testing	All modules	Completed
Validation Testing	Authentication, Registration fields	Completed
Negative Testing	All endpoints	Completed
Integration Testing	Auth flow, Course flow	Completed
System Testing	End-to-end flows	Completed
Regression Testing	Authentication module (V2)	Completed
Static / Design Review	SendCode, ValidateCode endpoints	Completed
Basic Security Testing	Protected endpoints (Auth headers)	Completed

7. Modules & Test Coverage

7.1 Authentication Module

Test Case	Type	Expected Result
Register with valid data	Functional	200 OK
Register with existing email	Negative	400 Bad Request
Register with invalid email format	Validation	400 Bad Request
Register with mismatched passwords	Validation	400 Bad Request

Test Case	Type	Expected Result
Signup with empty fields	Negative	400 Bad Request
Register with invalid phone number	Validation	400 Bad Request
Register with Email is null	Negative	400 Bad Request
Register with same phone and dadPhone	Negative	400 Bad Request
Student Login - valid credentials	Functional	200 OK
Login with unregistered account	Negative	401 Unauthorized
Login with unverified account	Negative	403 Forbidden
Login with invalid password	Negative	401 Unauthorized
Blocked user login	Negative	401 Unauthorized
Login with empty fields	Negative	400 Bad Request
Login with invalid email format	Negative	401 Unauthorized

7.2 Verification Module

Test Case	Type	Expected Result
Send code to valid unverified email	Functional	200 OK
Send code to already verified email	Negative	400 Bad Request
Send code to unregistered email	Negative	404 Not Found
Validate with valid code	Functional	200 OK
Validate with invalid/expired code	Negative	400 Bad Request

7.3 Course Management Module (Admin)

Test Case	Type	Expected Result
Admin valid login	Functional	200 OK
Admin login - wrong password	Negative	401 Unauthorized
Create course - valid data	Functional	200 OK
Get course by ID	Functional	200 OK
Get all courses	Functional	200 OK
Create course - no token	Security	401 Unauthorized
Update course by ID	Functional	200 OK
Update course - no token	Security	401 Unauthorized
Delete course by ID	Functional	200 OK

8. Entry & Exit Criteria

8.1 Entry Criteria

- API server is running and accessible on Local Development Environment
- Database is connected and seeded with required test data
- Postman environment variables are configured (Base_url, valid_email, authentication_token, etc.)
- Swagger documentation is available for reference

8.2 Exit Criteria

- All planned test cases have been executed
- All critical and high bugs are documented and reported to developers
- Review Findings Report is finalized
- Test Summary Report is prepared

9. Tools & Environment

Tool / Component	Details
API Testing Tool	Postman
Server	Local Development Environment (.NET Core)
API Documentation	Swagger UI
Database	SQL Server
Bug Tracking	Bug Report Template (Excel)
Test Management	Postman Collections
Environment Variables	Postman Environment

10. Risks & Mitigations

Risk	Impact	Mitigation
Test data pollution in DB	Medium	Use random emails and periodic DB cleanup
Unauthenticated course endpoints	Critical	Escalated to developer immediately

Risk	Impact	Mitigation
Sensitive data in URL (query params)	High	Documented in Review Findings Report
Email delivery not testable via API	Low	Manual verification using temporary mailbox